

(1001CJA102120045)

Test Pattern



# CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)  
PART TEST  
17-01-2021

## JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-I)

### ANSWER KEY

### PAPER-1

#### PART-1 : PHYSICS

| SECTION-I  | Q. | 1     | 2       | 3    | 4       | 5              | 6    | 7   | 8     | 9 | 10 |
|------------|----|-------|---------|------|---------|----------------|------|-----|-------|---|----|
|            | A. | A,C,D | A,B,C,D | B,C  | A,B,C,D | A,B,D          | A,C  | A,C | A,B,C | B | A  |
|            | Q. | 11    | 12      |      |         |                |      |     |       |   |    |
|            | A. | B     | A       |      |         |                |      |     |       |   |    |
| SECTION-II | Q. | 1     | 2       | 3    | 4       | 5              | 6    |     |       |   |    |
|            | A. | 0.02  | 2.00    | 0.81 | 0.25    | 99.63 to 99.65 | 0.15 |     |       |   |    |

#### PART-2 : CHEMISTRY

| SECTION-I  | Q. | 1       | 2     | 3     | 4     | 5    | 6    | 7   | 8     | 9 | 10 |
|------------|----|---------|-------|-------|-------|------|------|-----|-------|---|----|
|            | A. | A,B,C,D | A,B,D | A,C,D | B,C   | A,C  | A,B  | A,D | A,C,D | C | D  |
|            | Q. | 11      | 12    |       |       |      |      |     |       |   |    |
|            | A. | C       | D     |       |       |      |      |     |       |   |    |
| SECTION-II | Q. | 1       | 2     | 3     | 4     | 5    | 6    |     |       |   |    |
|            | A. | 1.00    | 4.58  | 9.40  | 13.60 | 7.00 | 9.25 |     |       |   |    |

#### PART-3 : MATHEMATICS

| SECTION-I  | Q. | 1    | 2    | 3            | 4    | 5    | 6    | 7   | 8   | 9 | 10 |
|------------|----|------|------|--------------|------|------|------|-----|-----|---|----|
|            | A. | A,C  | C,D  | B            | A,D  | A,D  | A    | B,D | A,C | C | D  |
|            | Q. | 11   | 12   |              |      |      |      |     |     |   |    |
|            | A. | B    | D    |              |      |      |      |     |     |   |    |
| SECTION-II | Q. | 1    | 2    | 3            | 4    | 5    | 6    |     |     |   |    |
|            | A. | 7.00 | 0.69 | 0.78 or 0.79 | 2.25 | 0.00 | 0.00 |     |     |   |    |

(1001CJA102120046)

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# CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Main)  
PART TEST  
17-01-2021

## JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-I)

### ANSWER KEY

### PAPER-2

#### PART-1 : PHYSICS

| SECTION-I  | Q. | 1     | 2    | 3    | 4    | 5            | 6    | 7    | 8    | 9      | 10   |
|------------|----|-------|------|------|------|--------------|------|------|------|--------|------|
|            | A. | D     | A    | C    | B    | B            | A    | A    | C    | A      | B    |
|            | Q. | 11    | 12   | 13   | 14   | 15           | 16   | 17   | 18   | 19     | 20   |
|            | A. | A     | D    | D    | C    | D            | A    | D    | A    | B      | B    |
| SECTION-II | Q. | 1     | 2    | 3    | 4    | 5            | 6    | 7    | 8    | 9      | 10   |
|            | A. | 30.00 | 3.05 | 3.00 | 0.50 | 6.40 to 6.41 | 0.80 | 0.75 | 6.00 | 395.00 | 4.00 |

#### PART-2 : CHEMISTRY

| SECTION-I  | Q. | 1    | 2    | 3    | 4    | 5    | 6    | 7    | 8    | 9     | 10    |
|------------|----|------|------|------|------|------|------|------|------|-------|-------|
|            | A. | D    | B    | B    | D    | D    | C    | D    | C    | C     | B     |
|            | Q. | 11   | 12   | 13   | 14   | 15   | 16   | 17   | 18   | 19    | 20    |
|            | A. | B    | A    | B    | C    | B    | C    | D    | C    | A     | C     |
| SECTION-II | Q. | 1    | 2    | 3    | 4    | 5    | 6    | 7    | 8    | 9     | 10    |
|            | A. | 1.00 | 5.00 | 1.00 | 6.00 | 6.50 | 5.00 | 9.02 | 2.00 | 11.00 | 10.00 |

#### PART-3 : MATHEMATICS

| SECTION-I  | Q. | 1    | 2    | 3            | 4    | 5    | 6    | 7    | 8    | 9     | 10   |
|------------|----|------|------|--------------|------|------|------|------|------|-------|------|
|            | A. | D    | A    | A            | C    | D    | C    | A    | C    | A     | D    |
|            | Q. | 11   | 12   | 13           | 14   | 15   | 16   | 17   | 18   | 19    | 20   |
|            | A. | A    | A    | B            | C    | B    | A    | B    | C    | C     | C    |
| SECTION-II | Q. | 1    | 2    | 3            | 4    | 5    | 6    | 7    | 8    | 9     | 10   |
|            | A. | 0.11 | 0.00 | 0.88 or 0.89 | 0.00 | 0.25 | 1.00 | 3.75 | 0.50 | 50.00 | 1.00 |



# CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)

PART TEST

17-01-2021

## JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-I)

### PAPER-1

#### PART-1 : PHYSICS

#### SOLUTION

##### SECTION-I

1. Ans. (A,C,D)

Sol. Given at  $t = 0$   $x = 0$   $\frac{\partial y}{\partial t} > 0$ ,  $y = 0$

$$y = A \sin(\omega t - kx + \theta)$$

$$0 = A \sin(0 - 0 + \theta).$$

$$\sin \theta = 0 \Rightarrow \theta = 0, \pi$$

$$\frac{\partial y}{\partial t} = A\omega \cos(\omega t - kx + \theta) = A\omega \cos \theta > 0$$

$$\theta = 0$$

Hence equation of wave

$$y = A \sin(\omega t - kx)$$

$$\text{Wave speed } C = \frac{\omega}{k}$$

$$k = \frac{2\pi}{\lambda}$$

2. Ans. (A,B,C,D)

Sol.  $V = \frac{\sqrt{F}}{\mu} = \frac{10}{(\mu_0 + \alpha x)^{1/2}} \Rightarrow \frac{dx}{dt} = \frac{10}{(\mu_0 + \alpha x)^{1/2}}$

$$\Rightarrow \int_0^{\ell} (\mu_0 + \alpha x)^{1/2} dx = \int_0^t 10 dt$$

$$t = \frac{1}{10} \times \int_0^{\ell} (\mu_0 + \alpha x)^{1/2} dx$$

3. Ans. (B,C)

Sol.  $\beta_2 - \beta_1 = 10 \left[ \log_{10} \frac{I_2}{I_0} - \log_{10} \frac{I_1}{I_0} \right]$

or  $30 = 10 \log_{10} \left( \frac{I_2}{I_1} \right)$

$$\therefore \frac{I_2}{I_1} = 10^3$$

$$\frac{P_2}{P_1} = \sqrt{1000} = 32$$

4. Ans. (A, B, C, D)

Sol. Since apparent frequency is constant for some time interval, hence their velocity or component of velocity along line of joining are constant.

5. Ans. (A,B,D)

Sol.  $\frac{dx_0}{D} = t(\mu - 1)$

$$\therefore x_0 = \frac{Dt(\mu - 1)}{d}$$

$$= \frac{1 \times 1.5 \times 10^{-7} \times 0.17}{3 \times 10^{-7}} = 8.5 \text{ cm}$$

6. Ans. (A,C)

Sol.  $-nKx + Kx = m \frac{d^2 x}{dt^2}$

$$-Kx = m \frac{d^2 (2x)}{dt^2}$$

$$\Rightarrow \frac{(n-1)K}{m} = \frac{K}{2m} \Rightarrow n = \frac{3}{2}$$

7. Ans. (A,C)

Sol.  $\frac{1}{2}KA^2 = \frac{1}{2}K\left(\frac{A}{2}\right)^2 + \frac{1}{2}mV^2$

$$\Rightarrow V = \sqrt{\frac{3K}{4m}}A$$

$$\frac{1}{2}KB^2 = \frac{1}{8}KA^2 + \frac{3}{16}KA^2$$

$$\Rightarrow B = \sqrt{\frac{5}{8}}A$$

8. Ans. (A,B,C)

Sol.  $x_1 = \frac{A}{2} \sin \omega t + \frac{\sqrt{3}A}{2} \cos \omega t$

$$= A \left[ \sin \omega t \cos \frac{\pi}{3} + \cos \omega t \sin \frac{\pi}{3} \right] = A \sin \left( \omega t + \frac{\pi}{3} \right)$$

$$x_2 = \sqrt{2}A \left[ \sin \omega t \cos \frac{\pi}{4} + \cos \omega t \sin \frac{\pi}{4} \right]$$

$$= \sqrt{2}A \sin \left( \omega t + \frac{\pi}{4} \right)$$

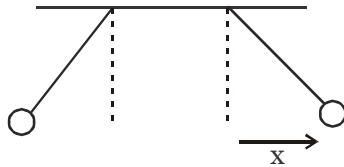
$$x = x_1 + x_2 = A \sin \left( \omega t + \frac{\pi}{3} \right) + \sqrt{2}A \sin \left( \omega t + \frac{\pi}{4} \right)$$

$$A_{\text{net}} = \sqrt{A^2 + (\sqrt{2}A)^2 + 2A\sqrt{2}A \cos\left(\frac{\pi}{3} - \frac{\pi}{4}\right)}$$

$$= \sqrt{3A^2 + 2\sqrt{2}A^2 \cos \frac{\pi}{12}}$$

9. Ans. (B)

Sol.



$$|\vec{F}| = mg \sin \theta + K(2x)$$

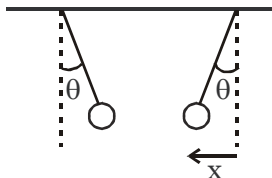
$$= mg \frac{x}{l} + \frac{mg}{l} \times 2x = \frac{3mgx}{l}$$

$$\text{T.E.} = mv_0^2$$

$$\text{P.E.} = \frac{3mgx^2}{2l}$$

10. Ans. (A)

Sol.



$$|\vec{F}| = mg \sin \theta = \frac{mgx}{l}$$

$$\text{T.E.} = mv_0^2$$

$$\text{P.E.} = \frac{mgx^2}{2l}$$

11. Ans. (B)

12. Ans. (A)

Sol. Let incident intensity is  $I_0$   
then reflected intensity is  $0.36 I_0$

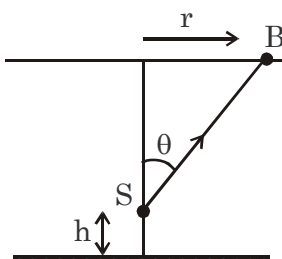
$$I_{\text{max}} = (\sqrt{I_0} + \sqrt{0.36 I_0})^2 = I_0 (1 + 0.6)^2$$

$$= (1.6)^2 I_0 = 2.56 I_0$$

$$I_{\text{min}} = (\sqrt{I_0} - \sqrt{0.36 I_0})^2 = I_0 (1 - 0.6)^2$$

$$= (0.4)^2 I_0 = 0.16 I_0$$

At point B phase difference between two waves is



$$\Delta x = 2h \cos \theta + \frac{\lambda}{2}$$

$$\Delta x = 2h \frac{D}{\sqrt{D^2 + r^2}} + \frac{\lambda}{2}$$

for first maxima ( $h = 2\lambda$ )

$$\Delta x = \lambda$$

$$\Rightarrow \lambda = \frac{\lambda}{2} + 2h \frac{D}{\sqrt{D^2 + r^2}}$$

$$\Rightarrow \frac{\lambda}{2} = 2h \frac{D}{\sqrt{D^2 + r^2}}$$

$$\Rightarrow \frac{\lambda}{2} = 2 \times 2\lambda \times \frac{D}{\sqrt{D^2 + r^2}}$$

$$\Rightarrow \frac{D^2}{D^2 + r^2} = \frac{1}{64}$$

$$\therefore D^2 + r^2 = 64 D^2$$

$$r = \sqrt{63} D$$

For Second maxima

$$\Delta x = 2\lambda$$

$$\therefore 2\lambda = \frac{\lambda}{2} + 2h \frac{D}{\sqrt{D^2 + r^2}}$$

$$\frac{3\lambda}{2} = 4\lambda \times \frac{D}{\sqrt{D^2 + r^2}}$$

$$\frac{9}{64} = \frac{D^2}{D^2 + r^2}$$

$$9r^2 + 9D^2 = 64D^2$$

$$9r^2 = 55D^2$$

$$r = \frac{\sqrt{55}}{3} D$$

$$h = 3\lambda$$

For second maxima

$$\Delta x = 2h \cos \theta + \frac{\lambda}{2} = 2\lambda$$

$$\Rightarrow 2 \times 3\lambda \times \frac{D}{\sqrt{D^2 + r^2}} = \frac{3\lambda}{2}$$

$$\Rightarrow \frac{D^2}{D^2 + r^2} = \frac{1}{16}$$

$$\Rightarrow D^2 + r^2 = 16D^2$$

$$\Rightarrow r = \sqrt{15} D$$

For first maxima

$$2h \cos \theta + \frac{\lambda}{2} = \lambda$$

$$2h \cos \theta = \frac{\lambda}{2}$$

$$2 \times 3 \times \frac{D}{\sqrt{D^2 + r^2}} = \frac{\lambda}{2}$$

$$\frac{D^2}{D^2 + r^2} = \frac{1}{144}$$

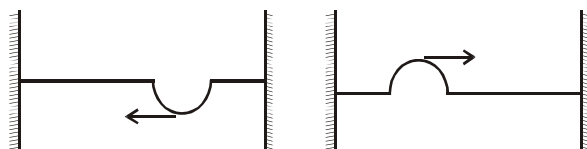
$$144D^2 = D^2 + r^2$$

$$r = \sqrt{143} D$$

## SECTION-II

1. Ans. 0.02

Sol. Given



$$\mu = 0.40 \text{ g/cm} = \frac{0.40}{1000} \times 100 = 0.040 \text{ kg/m}$$

Wave speed in the stretched string

$$v = \sqrt{\frac{F}{\mu}} = \sqrt{\frac{16}{0.040}} = 20 \text{ m/s}$$

The string will regain its shape after travelling the pulse a distance = 20 + 20 = 40 cm

$$\text{Thus time speed } t = \frac{0.40}{20} = 0.02 \text{ s}$$

2. Ans. 2.00

$$\text{Sol. } f = \frac{v}{40 + 2e} = \frac{3v}{4(31 + e)}$$

$$\Rightarrow \frac{3}{4}(40 + 2e) = 31 + e$$

$$\Rightarrow 120 + 6e = 124 + 4e$$

$$2e = 4$$

$$e = 2 \text{ cm}$$

3. Ans. 0.81

$$\text{Sol. } I_1 = \frac{0.64 I_0}{4} = 0.16 I_0$$

$$I_2 = \frac{I_0}{4} = 0.25 I_0$$

$$I_{\max} = (\sqrt{I_1} + \sqrt{I_2})^2 = (0.4\sqrt{I_0} + 0.5\sqrt{I_0})^2 = 0.81 I_0$$

4. Ans. 0.25

$$\text{Sol. } E = \frac{1}{2} I \omega_0^2$$

$$\frac{1}{4} M R^2 \omega_0^2 = 0.25 M R^2 \omega_0^2$$

5. Ans. 99.63 to 99.65

$$\text{Sol. } L = \frac{\lambda}{4n_2} = \frac{550}{4 \times 1.38} = 99.64$$

6. Ans. 0.15

$$\text{Sol. } d = 2a \alpha (n-1)$$

$$\beta = \frac{\lambda D}{d} = \frac{\lambda (a+b)}{2a \alpha (n-1)} = 0.15 \text{ cm}$$

$$\beta = \frac{\lambda (1+4)}{2 \times 1 \times \alpha (n-1)}$$

$$\beta = \frac{6000 \times 10^{-10} (1+4)}{2 \times 1 \times 2 \times 10^{-3} (1.5-1)}$$

$$= \frac{6000 \times 10^{-10} \times 5 \times 10}{4 \times 10^{-3} \times 15}$$

$$\frac{6}{4} \times 10^{-3}$$

$$\frac{1.5 \times 10^{-3} \times 100}{.15}$$

**PART-2 : CHEMISTRY**

**SOLUTION**

**SECTION-I**

1. Ans.(A,B,C,D)

2. Ans.(A,B,D)

3. Ans.(A,C,D)

Sol. (a)  $\text{Na}_2[\text{Be}(\text{OH})_4] + \text{H}_2 \uparrow$

(b)  $\text{Na}_2\text{CO}_3 + \text{H}_2\text{O}$

(c)  $\text{Li}_2\text{O} + \text{NO}_2 \uparrow + \text{O}_2 \uparrow$

(d)  $\text{NaOH} + \text{H}_2\text{O}_2 + \text{O}_2 \uparrow$

4. Ans.(B,C)

Sol. \* Rate constant doesn't depend upon concentration of reactant.

$$* t_{50\%} = \frac{0.693}{k} = \frac{0.693}{0.5} = 1.386 \text{ sec.}$$

5. Ans.(A,C)

Sol. (b)  $\text{Cr}^{3+} + \text{S} \downarrow$

(d)  $\text{CrO}_4^{2-}$  is tetrahedral

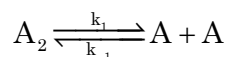
6. Ans.(A,B)

7. Ans.(A,D)

Sol. Rate law is written by slowest step.

$$R = k_2 [\text{A}][\text{B}_2] \quad \dots(4)$$

Since, 'A' is intermediate. We have to replace [A] in terms of reactant & product. from step



$$\frac{k_1}{k_{-1}} = \frac{[\text{A}]^2}{[\text{A}_2]}$$

$$[\text{A}] = \left[ \frac{k_1}{k_{-1}} [\text{A}_2] \right]^{\frac{1}{2}} \quad \dots(5)$$

from equation (4) & (5)

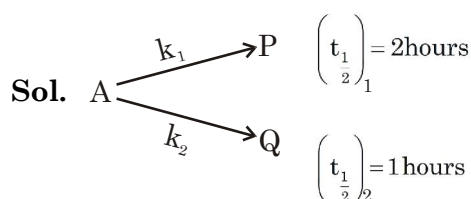
$$R = k_2 \left( \frac{k_1}{k_{-1}} \right)^{\frac{1}{2}} [\text{A}_2]^{\frac{1}{2}} [\text{B}_2]^1$$

The overall order of reaction is 1.5

order of reaction wrt  $\text{A}_2$  is 0.5

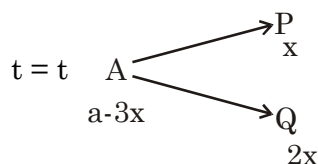
order of reaction wrt  $\text{B}_2$  is 1.0

8. Ans.(A,C,D)



$$\frac{k_1}{k_2} = \frac{\left( t_{\frac{1}{2}} \right)_2}{\left( t_{\frac{1}{2}} \right)_1} = \frac{1}{2} \left[ \begin{array}{l} \text{since } k_1 = \frac{\ln 2}{2} \\ k_2 = \frac{\ln 2}{1} \end{array} \right]$$

Suppose, initial moles of 'A' are 'a'



Given that  $\text{Q} = 2[\text{A}]_{\text{remaining}}$

$$2x = 2(a - 3x)$$

$$8x = 2a$$

$$x = \frac{2a}{8} = \frac{a}{4}$$

$$t = \frac{2.303}{k_1 + k_2} \log \frac{a}{a - 3x}$$

$$t = \frac{2.303}{k_1 + k_2} \log \frac{a}{a/4}$$

$$t = \frac{2.303}{k_1 + k_2} \log 4$$

$$t = \frac{2.303}{\frac{\ln 2}{2} + \frac{\ln 2}{1}} \log 4$$

$$t = \frac{2.303 \times 2 \log 2}{3 \frac{\ln 2}{2}}$$

$$t = \frac{4}{3} \text{ hours}$$

\* The amount of 'P' and amount of 'A' remaining will be equal after  $\frac{4}{3}$  hours.

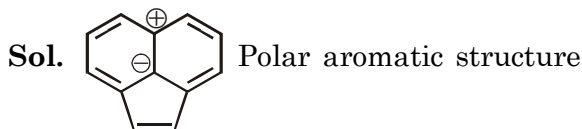
$$* \left( t_{\frac{1}{2}} \right)_{\text{overall}} = \frac{\left( t_{\frac{1}{2}} \right)_1 \times \left( t_{\frac{1}{2}} \right)_2}{\left( t_{\frac{1}{2}} \right)_1 + \left( t_{\frac{1}{2}} \right)_2}$$

$$= \frac{2 \times 1}{2 + 1} = \frac{2}{3} \text{ hours}$$

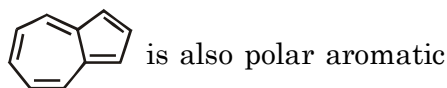
9. Ans.(C)

10. Ans.(D)

11. Ans.(C)



similarly



Compound because of 

12. Ans.(D)

### SECTION-II

1. Ans.(1.00)

Sol.  $k = Ae^{-(E_a/RT)}$

$$\ln k = \ln A - \frac{E_a}{RT}$$

$$y \text{ intercept} = \ln A = 33$$

$$A = e^{33}$$

$$\text{slope} = -\frac{E_a}{R} = -9.999 \times 10^3$$

Now, value of k at 30°C

$$k = A e^{-E_a/RT} = e^{33} \cdot e^{-\frac{9999}{303}} = e^{33} e^{-33}$$

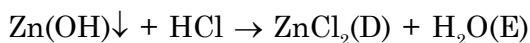
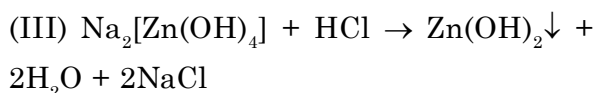
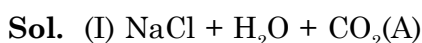
$$k = e^0$$

$$k = 1$$

2. Ans. (4.58)

Sol.  $x = \frac{4}{3}, y = \frac{3}{2}, z = \frac{7}{4}$

3. Ans.(9.40)



$$\text{So, } x = 44 + 32 + 18 = 94$$

4. Ans.(13.60)

Sol. Time period ( $T_n$ ) =  $1.5 \times 10^{-16} \frac{n^3}{z^2}$

$$T_{\text{H-atom}} = 2T_{\text{H-like atom}}$$

$$1.5 \times 10^{-16} \left( \frac{1^3}{1^2} \right) = 2[1.5 \times 10^{-16}] \times \frac{2^3}{z^2}$$

$$z^2 = 16$$

$$z = 4$$

separation energy in 3<sup>rd</sup> excited level is =

$$E_\infty - E_4 = -E_4$$

$$E_4 = -13.6 \frac{z^2}{n^2} \text{ eV}$$

$$= -13.6 \times \frac{4^2}{4^2}$$

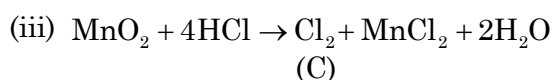
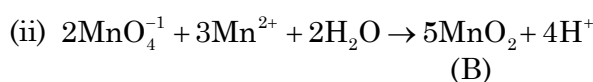
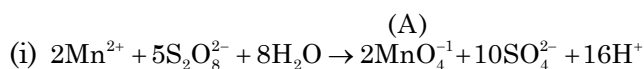
$$= -13.6 \text{ eV}$$

$$\text{ionization energy} = +13.60 \text{ eV}$$

5. Ans.(7.00)

6. Ans.(9.25)

Sol.



$$x = 3, y = 71$$

$$\Rightarrow (x + y) = 74$$

**PART-3 : MATHEMATICS**

**SOLUTION**

**SECTION-I**

1. Ans. (A,C)

**Sol.**  $\int \frac{7+6x^{13}}{\sqrt[3]{x^{17}-x^{30}}} dx = \int \frac{7x^{-8}+6x^5}{\sqrt[3]{x^{-7}-x^6}} dx$   
 put  $x^{-7}-x^6 = t^3$   
 $(-7x^{-8}-6x^5) dx = 3t^2 dt$   
 $(7x^{-8}+6x^5) dx = -3t^2 dt$   
 $\therefore I = \int \frac{-3t^2}{t} dt = -\int 3t dt = -\frac{3}{2} t^2 + C$   
 $= \frac{-3}{2} (x^{-7}-x^6)^{\frac{2}{3}} = -\frac{3}{2} \frac{(1-x^{13})^{\frac{2}{3}}}{x^{\frac{14}{3}}} + C$

2. Ans. (C,D)

**Sol.**  $\lim_{n \rightarrow \infty} \frac{n^{\frac{4041}{(2020)(2021)}} \sum_{r=1}^n [n^{2020} - r^{2020}]^{\frac{1}{2021}}}{\sum_{r=1}^n [n^{2021} - r^{2021}]^{\frac{1}{2021}}}$   
 $= \lim_{n \rightarrow \infty} \frac{n^{\frac{1}{2021}} n^{\frac{1}{2020}} \sum_{r=1}^n [n^{2020} - r^{2020}]^{\frac{1}{2021}}}{\sum_{r=1}^n [n^{2021} - r^{2021}]^{\frac{1}{2020}}}$   
 $= \frac{n^{\frac{1}{2021}} n^{\frac{1}{2020}} n^{\frac{2020}{2021}} \sum_{r=1}^n \left[ 1 - \left( \frac{r}{n} \right)^{2020} \right]^{\frac{1}{2021}}}{n^{\frac{2021}{2020}} \sum_{r=1}^n \left[ 1 - \left( \frac{r}{n} \right)^{2021} \right]^{\frac{1}{2020}}}$   
 $= \lim_{n \rightarrow \infty} \frac{\sum_{r=1}^n \left[ 1 - \left( \frac{r}{n} \right)^{2020} \right]^{\frac{1}{2021}}}{\sum_{r=1}^n \left[ 1 - \left( \frac{r}{n} \right)^{2021} \right]^{\frac{1}{2020}}}$   
 $= \lim_{n \rightarrow \infty} \frac{\frac{1}{n} \sum_{r=1}^n \left[ 1 - \left( \frac{r}{n} \right)^{2020} \right]^{\frac{1}{2021}}}{\frac{1}{n} \sum_{r=1}^n \left[ 1 - \left( \frac{r}{n} \right)^{2021} \right]^{\frac{1}{2020}}}$

$$\frac{\int_0^1 (1-x^{2020})^{\frac{1}{2021}} dx}{\int_0^1 (1-x^{2021})^{\frac{1}{2021}} dx}$$

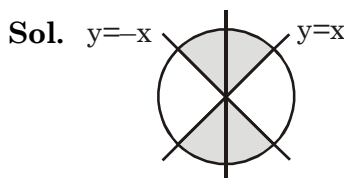
Let  $f(x) = (1-x^{2020})^{\frac{1}{2021}}$

$f^{-1}(x) = (1-x^{2021})^{\frac{1}{2020}}$

$$\int_0^1 (1-x^{2020})^{\frac{1}{2021}} + \int_1^0 (1-x^{2021})^{\frac{1}{2020}} = 0 - 0 = 0$$

$$\therefore \frac{\int_0^1 (1-x^{2020})^{\frac{1}{2021}}}{\int_0^1 (1-x^{2021})^{\frac{1}{2020}}} = 1$$

3. Ans. (B)



Area (T - S) =  $\frac{\pi(2)^2}{2} = 2\pi$

4. Ans. (A,D)

**Sol.**  $\frac{dy}{dx} + \frac{4x^3}{1+x^4} y = x$   
 $(1+x^4)dy + 4x^3 y dx = (x+x^5)dx$   
 $y(1+x^4) = \frac{x^2}{2} + \frac{x^6}{6} + C$   
 $y = \frac{\frac{x^2}{2} \left( 1 + \frac{x^4}{3} \right) + C}{1+x^4}$   
 $y = \frac{x^2(3+x^4)}{6(1+x^4)} + \frac{C}{1+x^4}$   
 $y = \frac{x^2}{3(1+x^4)} + \frac{x^2}{6} + \frac{C}{1+x^4}$

5. Ans. (A,D)

**Sol.**  $I = \int x^2 \ln x (x^3-1)^{-\frac{1}{2}} dx$   
 $II = \int \frac{x^2}{(x^3-1)^{\frac{1}{2}}} dx$  put  $x^3-1 = t^2$

$$3x^2 dx = 2t dt$$

$$= \frac{1}{3} \int \frac{2t}{t} dt = \frac{2}{3} \sqrt{x^3 - 1}$$

$$I = \ln x \cdot \frac{2}{3} \sqrt{x^3 - 1} - \frac{2}{3} \int \frac{\sqrt{x^3 - 1}}{x} dx$$

$$\Rightarrow 3 \int x^2 \ln x \cdot (x^3 - 1)^{-\frac{1}{2}} dx$$

$$= 2 \ln x \sqrt{x^3 - 1} - 2 \int \frac{\sqrt{x^3 - 1}}{x} dx$$

$$= 2 \ln x \sqrt{x^3 - 1}$$

6. Ans. (A)

$$\text{Sol. } I = \int_{\frac{\pi}{4}}^{\frac{9\pi}{4}} \frac{e^{\tan^{-1}(\cos x)}}{e^{\tan^{-1}(\cos x)} + e^{\tan^{-1}(\sin x)}} dx$$

$$= \int_0^{2\pi} \frac{e^{\tan^{-1}(\cos x)}}{e^{\tan^{-1}(\cos x)} + e^{\tan^{-1}(\sin x)}} dx$$

$$= \int_0^{\pi} \left[ \frac{e^{\tan^{-1}(\cos x)}}{e^{\tan^{-1}(\cos x)} + e^{\tan^{-1}(\sin x)}} + \frac{e^{\tan^{-1}(\cos x)}}{e^{\tan^{-1}(\cos x)} + e^{\tan^{-1}(\sin x)}} \right] dx$$

$$= \int_0^{\pi} \left[ \frac{e^{-\tan^{-1}(\cos x)}}{e^{-\tan^{-1}(\cos x)} + e^{\tan^{-1}(\sin x)}} + \int_0^{\pi} \frac{e^{\tan^{-1}(\cos x)}}{e^{\tan^{-1}(\cos x)} + e^{-\tan^{-1}(\sin x)}} dx \right]$$

$$= \int_0^{\pi} \left[ \frac{1}{1 + e^{\tan^{-1}(\sin x) + \tan^{-1}(\cos x)}} + \frac{e^{\tan^{-1}(\sin x) + \tan^{-1}(\cos x)}}{e^{\tan^{-1}(\sin x) + \tan^{-1}(\cos x)} + 1} \right] dx$$

$$= \int_0^{\pi} 1 dx = \pi$$

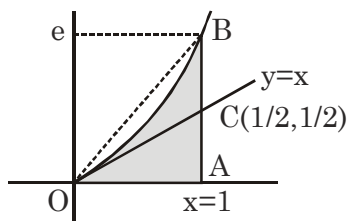
7. Ans. (B,D)

$$\text{Sol. } f(x) = xe^{\sqrt{x}}$$

$$f'(x) = \frac{xe^{\sqrt{x}}}{2\sqrt{x}} + e^{\sqrt{x}} = e^{\sqrt{x}} \left[ \frac{\sqrt{x}}{2} + 1 \right] > 0$$

$$f''(x) = e^{\sqrt{x}} \left[ \frac{1}{4\sqrt{x}} + \left( \frac{\sqrt{x}}{2} + 1 \right) \frac{1}{2\sqrt{x}} \right] > 0$$

$$f(x) = x \Rightarrow e^{\sqrt{x}} = 1$$



$$x = 0$$

$$f'(0) = 1$$

$$S = \int_0^1 xe^{\sqrt{x}} dx$$

$$S < \Delta OAB = \frac{1}{2} \cdot e$$

$$\text{If } x \in [0, 1] \quad \sqrt{x} > x$$

$$e^{\sqrt{x}} > e^x$$

$$\therefore xe^{\sqrt{x}} > xe^x$$

$$\int_0^1 xe^{\sqrt{x}} dx > \int_0^1 xe^x dx = [xe^x - e^x]_0^1$$

$$= e^1 - e^0 - [0 - 1] = 1$$

$$\int_0^1 xe^{\sqrt{x}} dx > 1$$

$$S > 1$$

8. Ans. (A,C)

$$\text{Sol. } \int dy = \int \frac{x^7 - 1}{x^2 \sqrt{2x^7 + 5}} dx$$

$$y = \frac{1}{5} \sqrt{2x^5 + \frac{5}{x^2}} + C$$

$$f(1) = \frac{\sqrt{7}}{5} + C = \frac{\sqrt{7}}{5} \Rightarrow C = 0$$

$$\therefore f(x) = \frac{1}{5} \sqrt{2x^5 + \frac{5}{x^2}}$$

$$f(-1) = \frac{\sqrt{3}}{5} = \frac{1.732}{5} = 0.32$$

$$\left[ I = \int \frac{x^7 - 1}{x^2 \sqrt{2x^7 + 5}} dx = \int \frac{x^7 - 1}{x^3 \sqrt{2x^5 + 5x^{-2}}} dx \right]$$

$$= \int \frac{x^4 - x^{-3}}{\sqrt{2x^5 + 5x^{-2}}} dx$$

$$\text{put } 2x^5 + 5x^{-2} = t^2 \Rightarrow (10x^4 - 10x^{-3})dx = 2t dt$$

$$\Rightarrow (x^4 - x^{-3})dx = \frac{t dt}{5}$$

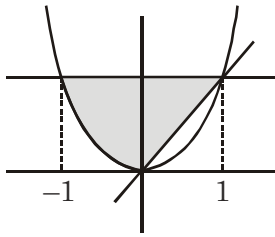
$$I = \int \frac{t dt}{5t} = \frac{1}{5} + t + C = \frac{1}{5} \sqrt{2x^5 + 5x^{-2}} + C$$

9. Ans. (C)

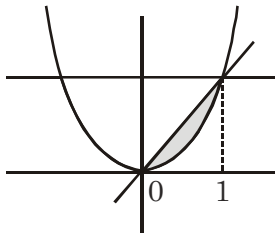
10. Ans. (D)

$$\text{Sol. } A = \int_{-1}^1 1 dx - \int_{-1}^0 x^2 dx - \frac{1}{2} \cdot 1 \cdot 1$$

$$= 2 - \frac{1}{2} - \left[ \frac{x^3}{3} \right]_{-1}^0 = \frac{3}{2} - \frac{1}{3} = \frac{7}{6}$$



$$B = \frac{1}{2} \cdot 1 \cdot 1 - \int_0^1 x^2 dx = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$$



11. Ans. (B)

12. Ans. (D)

**Sol.**  $f(x) = 2x + (1 + 40x)^{\frac{1}{60}}(1 + 18x)^{\frac{1}{54}}$

$$f'(x) = 2 + \frac{40}{60}(1 + 40x)^{-\frac{59}{60}}(1 + 18x)^{\frac{1}{54}} + (1 + 40x)^{\frac{1}{60}}(1 + 18x)^{-\frac{53}{54}} \cdot \frac{18}{54}$$

$$f'(0) = 2 + \frac{2}{3} + \frac{1}{3} = 3$$

$$\therefore e^x > 1 + x \text{ for } x > 0$$

$$\therefore \ln(1 + x) < x$$

$$\therefore \frac{1}{60} \ln(1 + 40x) + \frac{1}{54} \ln(1 + 18x) < \frac{2}{3}x + \frac{1}{3}x = x$$

$$\therefore (1 + 40x)^{\frac{1}{60}}(1 + 18x)^{\frac{1}{54}} < e^x$$

$$\therefore \int_0^1 f(x) dx < \int_0^1 (2x + e^x) dx = 1 + e - 1 = e$$

$$\int_{1/2}^{1/\sqrt{2}} x^{n-1} (\sin^{-1} x)^n dx$$

$$= \int_{\pi/6}^{\pi/4} (\sin t)^{n-1} t^n \cos t dt$$

$$= t^n \frac{\sin^n t}{n} \Big|_{\pi/6}^{\pi/4} - \int_{\pi/6}^{\pi/4} n t^{n-1} \frac{\sin^n t}{n} dt$$

$$= \frac{1}{n} \left[ \left( \frac{\pi}{4} \cdot \frac{1}{\sqrt{2}} \right)^n - \left( \frac{\pi}{6} \cdot \frac{1}{2} \right)^n \right] - \int_{\pi/6}^{\pi/4} t^{n-1} \sin^n t dt$$

$$I_n = \int_{1/2}^{1/\sqrt{2}} x^{n-1} (\sin^{-1} x)^n + \int_{\pi/6}^{\pi/4} x^{n-1} \sin^n x dx$$

$$= \frac{1}{n} \left[ \left( \frac{\pi}{4\sqrt{2}} \right)^n - \left( \frac{\pi}{12} \right)^n \right] = \frac{\pi^n}{4^n n} \left[ \left( \frac{1}{\sqrt{2}} \right)^n - \left( \frac{1}{3} \right)^n \right]$$

$$\frac{I_4}{(I_2)^2} = \frac{\frac{\pi^4}{4^4 \cdot 4} \left( \frac{1}{4} - \frac{1}{81} \right)}{\frac{\pi^4}{4^4 \cdot 4} \left( \frac{1}{2} - \frac{1}{9} \right)^2} = \frac{77}{4.81} \cdot \frac{81.4}{49} = \frac{11}{7}$$

$$\lim_{n \rightarrow \infty} (n^2 I_n) = \lim_{n \rightarrow \infty} \left\{ n \left( \frac{\pi}{4\sqrt{2}} \right)^n - n \left( \frac{\pi}{12} \right)^n \right\} = 0$$

## SECTION-II

1. Ans : 7.00

**Sol.**  $I = \int \frac{x^3 e^{2x} (x+2)}{\sqrt{1-x^4 e^{2x}}} dx = \int \frac{(x^2 e^x) x e^x (x+2)}{\sqrt{1-(x^2 e^x)^2}} dx$

$$\text{put } x^2 e^x = t \Rightarrow (2x e^x + x^2 e^x) dx = dt$$

$$\therefore x e^x (2+x) dx = dt \quad \therefore I = \int \frac{t dt}{\sqrt{1-t^2}} \quad 1-t^2 = z^2$$

$$= \int \frac{-z dz}{z} = -z + C = -\sqrt{1-t^2} + C$$

$$I = -\sqrt{1-x^4 e^{2x}} + C$$

$$a+b+c = 1+4+2 = 7$$

2. Ans : 0.69

**Sol.**  $x + y e^{y/x} = x e^{y/x} \frac{dy}{dx}$

$$\text{Let } y = 4x \Rightarrow \frac{dy}{dx} = u + x \frac{du}{dx}$$

$$\Rightarrow 1 + u e^u = e^u \left( u + x \frac{du}{dx} \right)$$

$$\Rightarrow \frac{dx}{x} = e^u du$$

$$\Rightarrow \ln x = e^u + c$$

$$y(1) = 0 \Rightarrow c = -1$$

$$\therefore \ln x = e^{y/x} - 1 \Rightarrow y = x \ln(1 + \ln x)$$

$$\frac{1}{e} y(e) = \ln(2) = 0.693$$

3. Ans : 0.78 or 0.79

**Sol.**  $I = \int_0^\infty \frac{1}{(1+X^{2021})(1+x^2)} dx$

$$= \int_0^1 \frac{dx}{(1+x^{2021})(1+x^2)} + \int_1^\infty \frac{1}{(1+x^{2021})(1+x^2)} dx$$

$$= \int_0^1 \frac{dx}{(1+x^{2021})(1+x^2)} + \int_1^0 \frac{1}{\frac{x^{2021}}{x^2} (x^2+1)} dx$$

$$= \int_0^1 \frac{dx}{(1+x^{2021})(1+x^2)} + \int_0^1 \frac{x^{2021}}{(1+x^{2021})(1+x^2)} dx$$

$$= \int_0^1 \frac{1}{1+x^2} dx = \frac{\pi}{4} = \frac{3.14}{4} = 0.78$$

4. Ans : 2.25

Sol.  $\int_0^a f(x) = g(a) \Rightarrow f(a) = g'(a)$

$$\int_a^5 f(x) = h(a) \Rightarrow -f(a) = h'(a)$$

$$g(a) + h(a) = 8, p(3) = 1 \Rightarrow g(3) = h(3)$$

$$\Rightarrow g(3) = h(3) = 4$$

$$p(a) = \frac{g(a)}{h(a)} \Rightarrow p'(a) = \frac{h(a)g'(a) - g(a)h'(a)}{h^2(a)}$$

$$a = 3 \Rightarrow \frac{9}{8} = \frac{h(3)f(3) + g(3)f(3)}{h(3)^2}$$

$$= \frac{f(3) \cdot [8]}{16} = \frac{9}{8} \Rightarrow f(3) = \frac{9}{4} = 2.25$$

5. Ans : 0.00

Sol.  $f''(x) \cdot \sin^2 x + 2f'(x) \sin 2x + 2f(x) \cdot \cos 2x = 6x$

$$f''(x) \cdot \sin^2(x) + f'(x) \cdot \sin 2x + f'(x) \cdot \sin 2x + 2f(x) \cos 2x = 6x$$

$$(f'(x) \cdot \sin^2 x)' + (f(x) \sin 2x)' = 6x$$

$$(f'(x) \cdot \sin^2 x + f(x) \cdot \sin 2x)' = 6x$$

$$((f(x) \cdot \sin^2 x)')' = 6x$$

$$(f(x) \cdot \sin^2(x))'' = 6x$$

$$(f(x) \cdot \sin^2(x))' = 3x^2 + c$$

$$f(x) \cdot \sin^2(x) = x^3 + cx + d$$

$$f(0) = 0 \Rightarrow d = 0$$

$$f(x) \sin^2 x = x^3 + cx$$

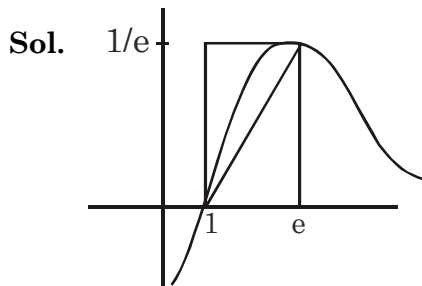
$$f(x) = \frac{x^3 + cx}{\sin^2 x}$$

$$\therefore f(x) \text{ is continuous at } x = 0$$

$$c = 0$$

$$f(x) = \frac{x^3}{\sin^2 x} \quad \therefore \left[ \frac{1}{f(1)} \right] = \left[ \sin^2 1 \right] = 0$$

6. Ans : 0.00



$$\frac{1}{2}(e-1) \frac{1}{e} < A < (e-1) \cdot \frac{1}{e}$$

$$\frac{1}{2} < \frac{eA}{e-1} < 1 \Rightarrow \left[ \frac{eA}{e-1} \right] = 0$$



# CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Main)

PART TEST

17-01-2021

## JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-I)

### PAPER-2

#### PART-1 : PHYSICS

#### SOLUTION

##### SECTION-I

1. Ans. (D)

Sol.  $A = 0.02 \cos(10\pi x)$ at  $x = 0.15$  m,  $A = 0$ at  $x = 0.3$  m,  $A = -0.02$  m

$$v = \frac{\omega}{K} = \frac{50\pi}{10\pi} = 5 \text{ m/s}$$

2. Ans. (A)

$$\text{Sol. } A_R = \left( \frac{v_2 - v_1}{v_1 + v_2} \right) A_i = \frac{A}{3}$$

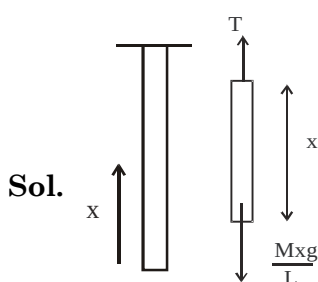
Let  $t = 0$ 

$$y_i = A \cos(kx)$$

$$y_R = -\frac{A}{3} \cos[k(x + vt)]$$

Reflection is from denser medium, so reflected wave is inverted.

3. Ans. (C)



Sol.

$$T = \frac{Mgx}{L}$$

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{gx}$$

 $t_1$  = time for lower half

$$v = \sqrt{gx} = \frac{dx}{dt}$$

$$\frac{L}{2} \int_0^{\frac{L}{2}} \frac{dx}{\sqrt{x}} = \sqrt{g} \int_0^t dt$$

$$2 \left[ \sqrt{x} \right]_0^{\frac{L}{2}} = \sqrt{g} t$$

$$2 \sqrt{\frac{L}{2}} = t_1 = \sqrt{\frac{2L}{g}}$$

Similarly for upper half,

$$2 \left[ \sqrt{x} \right]_{\frac{L}{2}}^L t_1 = \sqrt{g} t_2$$

$$2 \left( \sqrt{L} - \sqrt{\frac{L}{2}} \right) = \sqrt{g} t_2$$

$$2\sqrt{L} \left( \frac{\sqrt{2}-1}{\sqrt{2}} \right) = t_2$$

$$t_2 = \sqrt{\frac{2L}{g}} (\sqrt{2}-1)$$

4. Ans. (B)

$$\text{Sol. } \ell_1 + 0.3d = \frac{\lambda}{3}, \ell_2 + 0.3d = \frac{3\lambda}{4}$$

$$\frac{\lambda}{2} = \ell_2 - \ell_1 = 0.25 \text{ m or } \frac{\lambda}{4} = 0.125 \text{ m}$$

$$0.3d = \frac{\lambda}{4} - \ell_1 = 0.025 \text{ m}$$

5. Ans. (B)

$$\text{Sol. } dp = -\frac{Bds}{dx}$$

$$\Delta P = -157000 \text{ Pa} \cos(10\pi x)$$

6. Ans. (A)

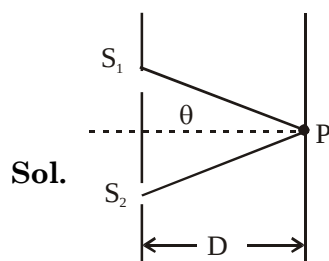
Sol.  $\lambda_1 = UT + \frac{UT}{20} = \frac{U}{f} + \frac{U}{20f}$

$$\lambda_1 = \lambda_0 + \frac{\lambda_0}{20} = \frac{21}{20} \lambda_0$$

$$\lambda_2 = UT - \frac{UT}{20}$$

$$\lambda_2 = \lambda_0 - \frac{\lambda_0}{20} = \frac{19\lambda_0}{20}$$

7. Ans. (A)



$$\tan \theta \approx \theta = \frac{d}{D}$$

$$\text{and } \beta = \frac{\lambda D}{d} = \frac{\lambda}{\theta}$$

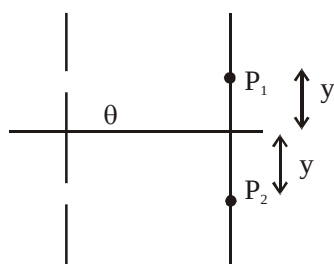
8. Ans. (C)

Sol.  $I = I_{\max} \cos^2 \left( \frac{\Delta \phi}{2} \right) = \frac{I_{\max}}{4}$

$$\cos \left( \frac{\Delta \phi}{2} \right) = \frac{1}{2}$$

$$\Delta \phi = \frac{2\pi}{3} = \frac{2\pi}{\lambda} \Delta x$$

$$\Delta x = d \cos \theta \approx d \tan \theta = \frac{dy}{D}$$



$$\frac{dy}{D} = \frac{\lambda}{3}$$

$$y = \frac{\lambda D}{3d} = \frac{\beta}{3}$$

9. Ans. (A)

Sol.  $\frac{dv}{dt} = 2t$

$$v = t^2$$

$$\text{Now, } y = \pm \frac{nD\lambda}{d}$$

$$v_y = \pm \frac{n\lambda}{d} \left( \frac{dD}{dt} \right)$$

$$v_y = \pm \frac{n\lambda}{d} t^2$$

Thus, velocity of  $n^{\text{th}}$  maxima

$$\vec{v}_n = 4\hat{i} \pm \frac{n\lambda}{d} t^2 \hat{j}$$

At  $t = 2$  sec

$$\vec{v}_n = 4\hat{i} \pm \frac{4n\lambda}{d} \hat{j}$$

10. Ans. (B)

Sol. Let  $a$  be the width of each slit. Linear separation between 10 bright fringes.

$$x = 5\beta = \frac{5\lambda D}{d}$$

corresponding angular separation

$$\theta_1 = \frac{x}{D} = \frac{5\lambda}{d}$$

Now, the angular width of central maximum in the diffraction pattern of a single slit,

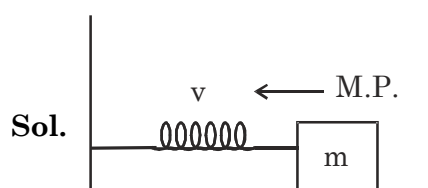
$$\theta_2 = \frac{2\lambda}{a}$$

As  $\theta_2 = \theta_1$

$$\frac{2\lambda}{a} = \frac{5\lambda}{d}$$

$$\text{or } a = 2d/5 = 2.00/5 \text{ mm} = 0.4 \text{ mm}$$

11. Ans. (A)

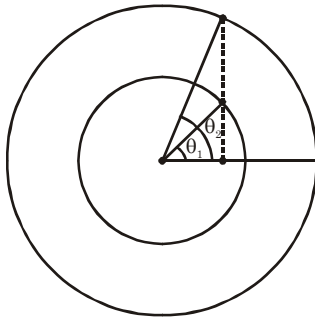


$$x = -a \sin \omega t, v = -b \cos \omega t$$

$$\text{or } x = A \sin \omega t, v = B \cos \omega t$$

12. Ans. (D)

Sol.  $\cos \theta_2 = \frac{2}{5}$ ,



$$\cos \theta_1 = \frac{4}{5}$$

$$\Delta \phi = -\cos^{-1}\left(\frac{4}{5}\right) + \cos^{-1}\left(\frac{2}{5}\right)$$

13. Ans. (D)

Sol.  $U = 5x^2 - 20x \Rightarrow F = \frac{-dU}{dx} = -10(x - 2)$

M.P. at  $x = 2\text{m}$

14. Ans. (C)

Sol.  $4v^2 = 25 - x^2$

Differentiating,  $v = \frac{dv}{dx} \times 8 = -2x$

$$\Rightarrow \omega^2 = \frac{1}{4}$$

$$v_{\max} = \frac{5}{2}$$

$$4 \times 2v \frac{dv}{dx} = -2x$$

$$v^2 = \frac{1}{4}(25 - x^2)$$

$$a = 5$$

15. Ans. (D)

Sol.  $A = A_0 e^{-rt}$

$$\frac{A_0}{2} = A_0 e^{-r(10T)}$$

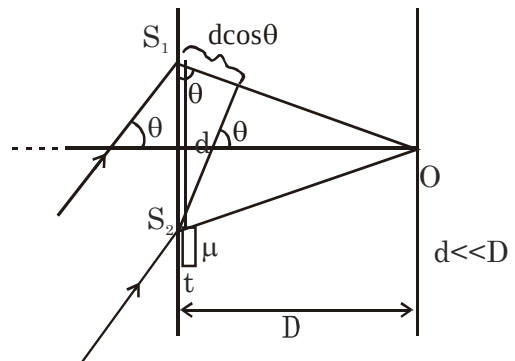
$$10rT = \ln 2 \dots (1)$$

$$\frac{A_0}{1000} = A_0 e^{-rt}, rt = \ln 1000 \dots (2)$$

$$\frac{rt}{10rT} = \frac{2.3 \times 3}{0.693} \approx 10$$

$$t = 100T = 100 \times 0.2 = 20\text{s}$$

16. Ans. (A)



Sol.

$$d \cos \theta = (\mu - 1)t$$

17. Ans. (D)

Sol.  $\vec{S} = \frac{\vec{E} \times \vec{B}}{\mu_0}$

is called the Poynting vector after its discoverer, John Poynting. The average magnitude of  $\vec{S}$  is the intensity of the wave, and the direction of  $\vec{S}$  is the direction of propagation of the wave.

18. Ans. (A)

Sol.  $I^{\text{st}}$  minima

$$\theta = 30$$

$$\sin \theta = \frac{n\lambda}{a}$$

$$\frac{1}{2} = \frac{1 \times \lambda}{1\mu\text{m}}$$

$$\lambda = 0.5 \mu\text{m}$$

$$D = 0.5 \text{ m}$$

$$\beta = 1 \times 10^{-2} \text{ m}$$

$$d = ?$$

$$\beta = \frac{D\lambda}{d}$$

$$10^{-2} = \frac{1 \times 0.5 \times 10^{-6}}{2d}$$

$$d = \frac{0.25 \times 10^{-6}}{10^{-2}}$$

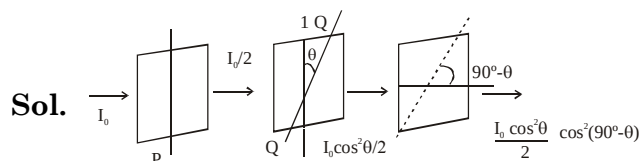
$$d = 0.25 \times 10^{-4} \text{ m}$$

$$d = 25 \times 10^{-6} \text{ m}$$

$$d = 25 \mu\text{m}$$

**ALLEN**

 19. **Ans. (B)**
**Sol.** C will vibrate with max amplitude as its frequency is same as that of A.

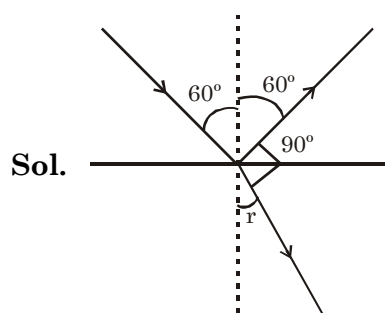
 20. **Ans. (B)**


$$I_R = \frac{I_0}{2} \cos^2 \theta \sin^2 \theta$$

$$I_R = \frac{I_0}{8} (\sin 2\theta)^2$$

$$(I_R)_{\max} = \frac{I_0}{8} \text{ at } \theta = \frac{\pi}{4}$$

### SECTION-II

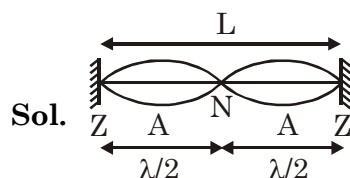
 1. **Ans. 30.00**


At polarizing angle (Brewster's angle) angle between reflected and refracted ray is  $90^\circ$

 2. **Ans. 3.05**
**Sol.** Limit of resolution of telescope

$$= \Delta\theta = \frac{1.22\lambda}{a}$$

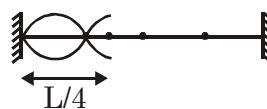
$$\theta = 3.05 \text{ rad}$$

 3. **Ans. 3.00**


$$\frac{\lambda}{2} + \frac{\lambda}{2} = L \Rightarrow \lambda = L$$

$$f = \frac{v}{\lambda} \Rightarrow f = \frac{v}{L} = 100 \text{ (Given)}$$

Next mode



$$\frac{\lambda}{2} + \frac{\lambda}{4} = \frac{L}{4} \Rightarrow \frac{3\lambda}{4} = \frac{L}{4}$$

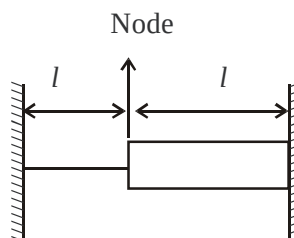
$$\Rightarrow \lambda = \frac{L}{3}$$

So, frequency,  $f = \frac{v}{\lambda}$

$$f = \frac{3v}{L} = 300 \text{ Hz}$$

 4. **Ans. 0.50**

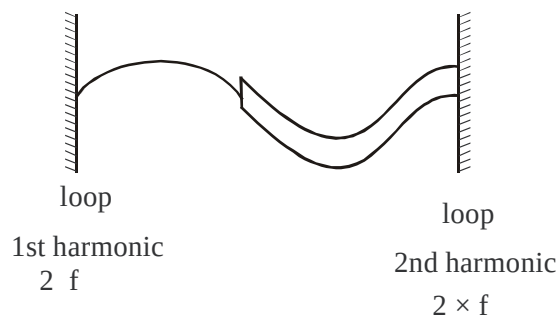
**Sol.**  $v = \sqrt{\frac{T}{u}} = \frac{\sqrt{TL}}{m} = \sqrt{\frac{T}{\rho A}} = \sqrt{\frac{T}{\rho \pi r^2}}$



$$v_1 \propto \frac{1}{r}, v_2 \propto \frac{1}{2r}$$

$$\frac{v_1}{v_2} = 2$$

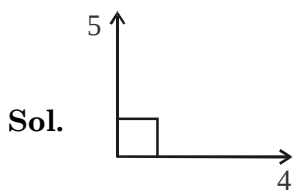
$f_1$  = fundamental frequency for wire 1



$$f_1 = \frac{v_1}{2l} = \left( \frac{2v_2}{2l} \right) = 2f$$

$$\text{Let } f = f_2 = \frac{v_2}{2l}$$

5. Ans. 6.40 to 6.41



$$\text{Area} = \sqrt{4^2 + 5^2 + 2 \times 4 \times 5 \times \cos \frac{\pi}{2}} = \sqrt{41} \approx 6.4$$

6. Ans. 0.80

Sol. Let  $I_1 = I$

$$\& I_2 = 4I$$

$$\text{then } I_{\max} = (\sqrt{I} + \sqrt{4I})^2 = 9I$$

$$\& I_{\min} = (\sqrt{4I} - \sqrt{I})^2 = I$$

$$\frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} = \frac{9I - I}{9I + I} = 0.8$$

7. Ans. 0.75

Sol.  $\Delta\phi = \frac{2\pi}{\lambda}(\Delta x) = \frac{2\pi}{\lambda}\left(\frac{\lambda}{6}\right) = \frac{\pi}{3}$

$$I = 4I_0 \cos^2\left(\frac{\Delta\phi}{2}\right) = 4I_0 \cos^2\left(\frac{\pi}{6}\right)$$

$$= 4I_0 \times \frac{3}{4} = 3I_0$$

$$\text{Now } \frac{I}{4I_0} = \frac{3}{4} = 0.75$$

8. Ans. 6.00

Sol. Tension in the wire

$$= 2\left(\frac{2m_1 m_2}{m_1 + m_2}\right)g$$

$$= 2\left(\frac{2 \times m \times 2m}{m + 2m}\right)g$$

$$T = \frac{8mg}{3} \dots\dots(i)$$

for air column :

$$\frac{x}{2} = \frac{\lambda}{4} \Rightarrow \lambda = 2x$$

$$f_a = \frac{v}{\lambda} = \frac{400}{2x} = \frac{200}{x} \& f_w = \frac{u}{2x} = \frac{1}{2x} \sqrt{\frac{T}{\mu}}$$

$$\text{Now, } f_a = f_w$$

$$\frac{200}{x} = \frac{1}{2x} \sqrt{\frac{T}{\mu}}$$

$$\sqrt{\frac{T}{\mu}} = 400$$

$$T = \mu \times 16 \times 10^4$$

$$\frac{8mg}{3} = 10^{-4} \times 16 \times 10^4$$

$$mg = 6$$

$$m = 6 \times 10^{-1} \text{ kg}$$

9. Ans. 395.00

Sol. Frequency in fundamental mode

$$f_0 = \frac{1}{2\ell} \sqrt{\frac{T}{\mu}}$$

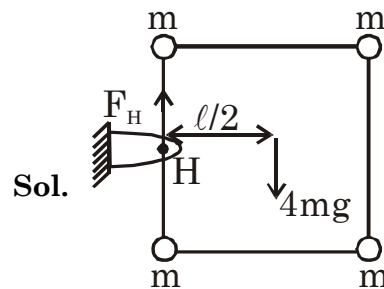
T decreases with increase in temperature

$\Rightarrow f_0$  will also decrease

$$\text{Now, } 400 - f_0 = 5$$

$$f_0 = 395 \text{ Hz}$$

10. Ans. 4.00



$$\tau = 4mg \frac{\ell}{2}$$

$$I_H \alpha = 4mg \frac{\ell}{2}$$

$$\left\{ 4m \left( \frac{\ell}{\sqrt{2}} \right)^2 + 4m \left( \frac{\ell}{2} \right)^2 \right\} \alpha = 2mg\ell$$

$$3m\ell^2 \alpha = 2mg\ell$$

$$\alpha = \frac{2g}{3\ell} \dots\dots(i)$$

$$4mg - F_H = 4m \left( \frac{\ell}{2} \alpha \right)$$

$$F_H = 4mg - 2m\ell \times \frac{2g}{3\ell}$$

$$F_H = 4mg - \frac{4mg}{3} = \frac{8mg}{3} = \frac{8 \times 0.15 \times 10}{3}$$

$$F_H = 3.92 \text{ N}$$

or 4N (if g is taken 10 m/s<sup>2</sup>)

**PART-2 : CHEMISTRY**

**SOLUTION**

**SECTION-I**

1. **Ans.(D)**

**Sol.**  $r \propto [P]^1$  for first order  $t_{75\%} = 2(t_{50\%})$   
 $r \propto [Q]^0$  for zero order  $A_t = A_0 - kt$   
 $r = k[P]^1[Q]^0$

2. **Ans.(B)**

3. **Ans.(B)**

4. **Ans. (D)**

**Sol.**  ${}_0n^1 + {}_{92}^{235}\text{U} \rightarrow {}_{54}^{142}\text{Xe} + {}_{38}^{90}\text{Sr} + 4{}_0n^1$

5. **Ans.(D)**

6. **Ans.(C)**

7. **Ans.(D)**

8. **Ans.(C)**

9. **Ans.(C)**

10. **Ans.(B)**

**Sol.**  $\frac{1}{2}mv^2 = hv - hv_0$

$$\Rightarrow v \propto [v - v_0]^{\frac{1}{2}}$$

$$\lambda = \frac{h}{mv}$$

Acc. to de-Broglie wave equation :

$$\Rightarrow \lambda \propto \frac{1}{(v - v_0)^{\frac{1}{2}}}$$

11. **Ans.(B)**

12. **Ans.(A)**

13. **Ans.(B)**

**Sol.**  $2\pi r_n = n\lambda$

$$2\pi r_1 \times \frac{n^2}{Z} = n\lambda$$

$$2\pi a_0 \times \frac{4}{1} = 2 \times \lambda$$

$$\Rightarrow \lambda = 4\pi a_0$$

14. **Ans.(C)**

15. **Ans.(B)**

**Sol.**  $(\text{NH}_4)_2\text{Cr}_2\text{O}_7 \xrightarrow{\Delta} \text{N}_2 \uparrow + \underset{\text{green residue}}{\text{Cr}_2\text{O}_3} + 4\text{H}_2\text{O}$

16. **Ans.(C)**

**Sol.** Total numbers of peaks in radial probability distribution function curve =  $n - l$

17. **Ans.(D)**

18. **Ans.(C)**

19. **Ans.(A)**

**Sol.**  $r_1 = k A_0$

$$r_2 = k A_t \quad t_{1/2} = \frac{0.693}{\lambda}$$

$$\Rightarrow \frac{r_1}{r_2} = \frac{A_0}{A_t} = \frac{15.2}{7.6}$$

$$\lambda \times t = \log_e \frac{15.2}{7.6}$$

$$t = \frac{1}{\lambda} \log_e \frac{15.2}{7.6} = \frac{5760}{0.693} \times \log_e \frac{15.2}{7.6}$$

$$= 5760$$

20. **Ans.(C)**

**SECTION-II**

1. **Ans. (1.00)**

**Sol.** Only  $\text{Na}_2\text{S}_2\text{O}_3$

2. **Ans.(5.00)**

**Sol.**  $r = -\frac{1}{2} \frac{dA}{dt} = k[A]^1[B]^0$

$$6.93 \times 10^{-3} = k (0.10)$$

$$k = 6.93 \times 10^{-2}$$

$t$  = required to consume

$$\text{half of A} = \frac{0.693}{2 \times 6.93 \times 10^{-2}} = 5$$

3. **Ans.(1.00)**

4. **Ans.(6.00)**

5. **Ans.(6.50)**

**Sol.**  $R = R_0(A)^{\frac{1}{3}}$

$$R = 1.3 \times 10^{-15} \times (125)^{\frac{1}{3}}$$

$$= 5 \times 1.3 \times 10^{-15} \text{ m}$$

$$= 6.5 \times 10^{-15} \text{ m}$$

6. **Ans.(5.00)**

(a), (b), (d), (e), (g)

7. **Ans. (9.02)**

$$x = 9$$

$$y = 2$$

8. **Ans. (2.00)**

$$2 = \frac{hc}{\lambda} - \phi \quad \dots\dots\dots\text{(i)}$$

$$6 = \frac{2hc}{\lambda} - \phi \quad \dots\dots\dots\text{(ii)}$$

Solving :  $\phi = 2\text{eV}$  Ans.

9. **Ans.(11.00)**

10. **Ans.(10.00)**

**PART-3 : MATHEMATICS**

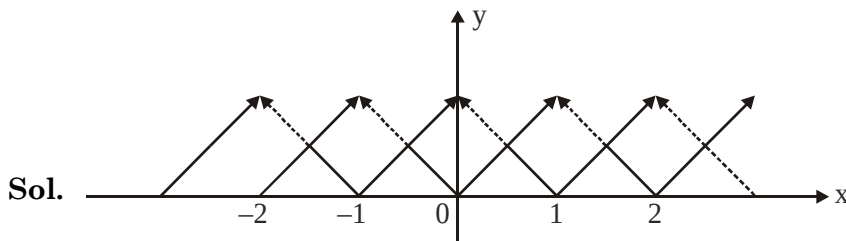
**SOLUTION**

**SECTION-I**

1. **Ans. (D)**

**Sol.**  $I = \int \frac{\sqrt{x-1}}{x\sqrt{x+1}} dx = \int \frac{x-1}{x\sqrt{x^2-1}} dx = \int \frac{dx}{\sqrt{x^2-1}} - \int \frac{dx}{x\sqrt{x^2-1}} = \ln|x + \sqrt{x^2+1}| - \sec^{-1} x + c$

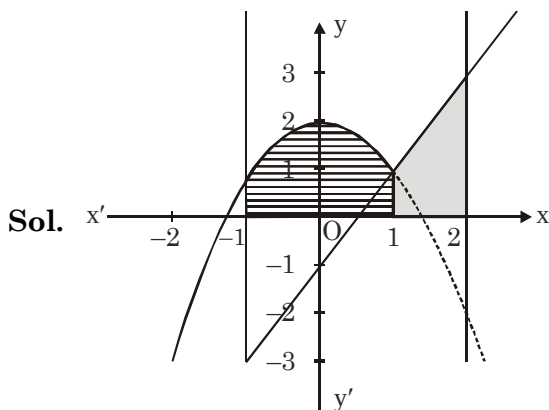
2. **Ans. (A)**



The graph with solid line is the graph  $f(x) = \{x\}$  and the graph with dotted lines is the graph of  $f(x) = \{-x\}$ . Now, the graph of  $\min(\{x\}, \{-x\})$  is the graph with dark solid lines.

$$\int_{-100}^{100} f(x) dx = \text{Area of 200 triangles shown as solid dark lines in the diagram} = 200 \cdot \frac{1}{2} (1) \left( \frac{1}{2} \right) = 50$$

3. **Ans. (A)**



$$A = \int_{-1}^1 (-x^2 + 2) dx + \int_1^2 (2x - 1) dx = \left( -\frac{x^3}{3} + 2x \right)_{-1}^1 + (x^2 - x)_1^2 = \frac{16}{3} \text{ sq. units}$$

4. **Ans. (C)**

**Sol.** We have  $\frac{dy}{dx} = (e^y - x)^{-1} \Rightarrow \frac{dx}{dy} = e^y - x \Rightarrow \frac{dx}{dy} + x = e^y$ ;

So I.F. =  $e^{\int dy} = e^y \quad \therefore$  General solution is given by

$$xe^y = \frac{1}{2} e^{2y} + C \Rightarrow x = \frac{e^y}{2} + Ce^{-y}$$

As  $y(0) = 0$ , so  $C = \frac{-1}{2} \quad \therefore x = \frac{e^y}{2} - \frac{1}{2} e^{-y}$

$$\Rightarrow e^y - e^{-y} = 2x \Rightarrow e^{2y} - 2xe^y - 1 = 0 \Rightarrow 2e^y = 2x \pm \sqrt{4x^2 + 4}$$

But  $e^y = x - \sqrt{x^2 + 1}$  (Rejected)

Hence  $y = \log_e (x + \sqrt{x^2 + 1})$

5. Ans. (D)

$$\begin{aligned}
 \text{Sol. } \int x \log \left( 1 + \frac{1}{x} \right) dx &= \int x \log(x+1) dx - \int x \log x dx \\
 &= \frac{x^2}{2} \log(x+1) - \frac{1}{2} \int \frac{x^2}{x+1} dx - \frac{x^2}{2} \log x + \frac{1}{2} \int \frac{x^2}{x} dx + C \\
 &= \frac{x^2}{2} \log(x+1) - \frac{1}{2} \int \left( x - 1 + \frac{1}{x+1} \right) dx - \frac{x^2}{2} \log x + \frac{1}{4} x^2 + C \\
 &= \frac{x^2}{2} \log(x+1) - \frac{x^2}{2} \log x - \frac{1}{2} \left( \frac{x^2}{2} - x \right) - \frac{1}{2} \log(x+1) + \frac{1}{4} x^2 + C \\
 &= \frac{x^2}{2} \log(x+1) - \frac{x^2}{2} \log x - \frac{1}{2} \log(x+1) + \frac{1}{2} x + C
 \end{aligned}$$

$$\text{Hence, } A = \frac{1}{2}$$

6. Ans. (C)

$$\begin{aligned}
 \text{Sol. Let } A &= \lim_{n \rightarrow \infty} \left[ \tan \frac{\pi}{2n} \tan \frac{2\pi}{2n} \dots \tan \frac{n\pi}{2n} \right]^{\frac{1}{n}} \\
 \therefore \log A &= \lim_{n \rightarrow \infty} \frac{1}{n} \left[ \log \tan \frac{\pi}{2n} + \log \tan \frac{2\pi}{2n} + \dots + \log \tan \frac{n\pi}{2n} \right] \\
 &= \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{n} \log \tan \frac{\pi r}{2n} = \int_0^1 \log \tan \left( \frac{\pi}{2} x \right) dx = \frac{2}{\pi} \int_0^{\pi/2} \log \tan y dy \quad \dots(1)
 \end{aligned}$$

$$\left[ \text{Putting } \frac{1}{2} \pi x = y \text{ and } dx = \left( \frac{2}{\pi} \right) dy \right]$$

$$\begin{aligned}
 \text{Now, let } I &= \int_0^{\pi/2} \log \tan y dy \quad \therefore I = \int_0^{\pi/2} \log \tan \left( \frac{1}{2} \pi - y \right) dy \\
 &= \int_0^{\pi/2} \log \cot y dy = - \int_0^{\pi/2} \log \tan y dy = -1 \text{ or } I + I = 0 \text{ or } 2I = 0 \text{ or } I = 0
 \end{aligned}$$

Thus, from equation (1),  $\log A = 0$  or  $A = e^0 = 1$ .

7. Ans. (A)

Sol. Curve tracing :  $y = xe^x$

$$\text{Let } \frac{dy}{dx} = 0 \Rightarrow e^x + xe^x = 0 \text{ or } x = -1.$$

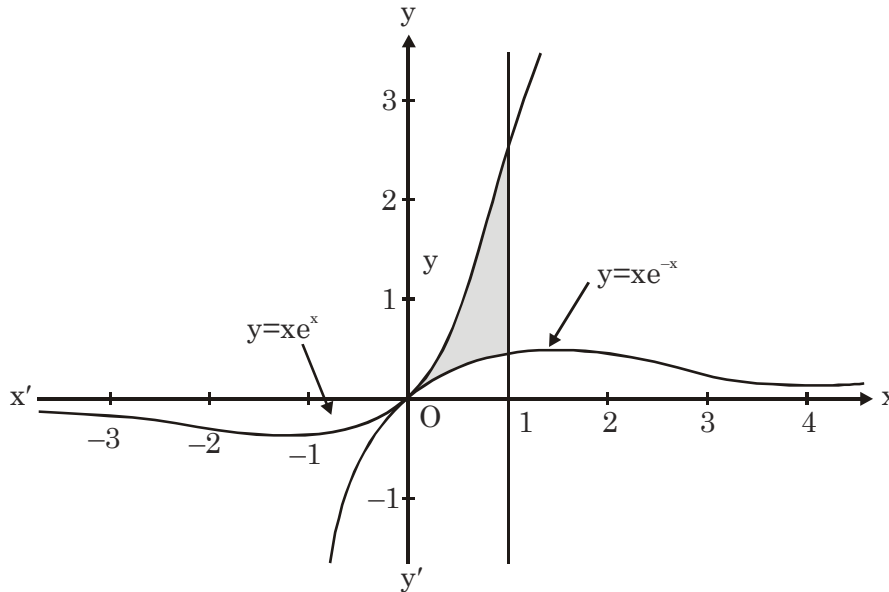
Also, at  $x = -1$ ,  $\frac{dy}{dx}$  changes sign from -ve to +ve,

Hence,  $x = -1$  is a point of minima.

When  $x \rightarrow \infty$ ,  $y \rightarrow \infty$

$$\text{Also } \lim_{x \rightarrow \infty} x e^x = \lim_{x \rightarrow \infty} \frac{x}{e^{-x}} = \lim_{x \rightarrow \infty} \frac{1}{e^{-x}} = 0$$

With similar types of arguments, we can draw the graph of  $y = x e^{-x}$ .



$$\text{Required area} = \int_0^1 x e^x dx - \int_0^1 x e^{-x} dx = [x e^x]_0^1 - \int_0^1 e^x dx - \left( [-x e^{-x}]_0^1 + \int_0^1 e^{-x} dx \right)$$

$$= e - (e - 1) - (-e^{-1} - (e^{-1} - 1)) = \frac{2}{e} \text{ sq. units}$$

8. Ans. (C)

Sol.  $\frac{dy}{dx} = \frac{y}{x} \left[ \log \frac{y}{x} + 1 \right]$

Put  $y = vx$

$$v + x \frac{dv}{dx} = v \log v + v$$

$$\therefore \frac{dv}{v \log v} = \frac{dx}{x}$$

$$\therefore \log (\log v) = \log x + \log c = \log cx$$

$$\therefore \log \frac{y}{x} = \log cx$$

9. Ans. (A)

Sol.  $R \times (P' \cup Q)' = R \times (P \cap Q) = (R \times P) \cap (R \times Q)$

10. Ans. (D)

Sol. (2, 3) belongs to R but (3, 2) is not  $\Rightarrow$  R is not symmetric

11. Ans. (A)

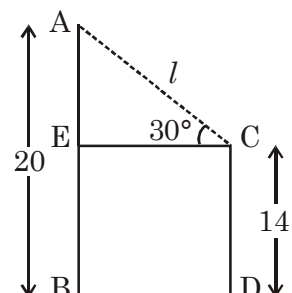
Sol. Let  $AB = 20$  and  $CD = 14$  be the two poles.

$AC = l$  is the length of the wire.

CE, the horizontal through C,  $\angle ACE = 30^\circ$

$$\frac{AE}{l} = \sin 30^\circ = \frac{1}{2}$$

$$\therefore l = 2AE = 12\text{m}$$



12. Ans. (A)

Sol. Let  $p(n) : 2^n < n!$

Then,  $p(1) : 2! < 1!$ , which is not true

$p(2) : 2^2 < 2!$ , which is not true

$p(3) : 2^3 < 3!$ , which is not true

$p(4) : 2^4 < 4!$ , which is true.

Let  $p(k)$  be true if  $k \geq 4$ , i.e.,  $2^k < k!$ ,  $k \geq 4$

$\Rightarrow 2 \cdot 2^k < 2(k!) \Rightarrow 2^{k+1} < k(k!) (\because k \geq 4 > 2)$

$\Rightarrow 2^{k+1}(k+1)! \Rightarrow p(k+1)$  is true.

Hence, we conclude that  $p(n)$  is not true for  $n = 2, 3$  but hold true for  $n \geq 4$ .

13. Ans. (B)

Sol.  $p \rightarrow (q \rightarrow p) = \sim p \vee (q \rightarrow p) = \sim p \vee (\sim q \vee p)$  since  $p \vee \sim p$  is always true  
 $= \sim p \vee p \vee q = p \rightarrow (p \vee q)$

14. Ans. (C)

Sol. Here mean  $A = 5$ .

Let the frequency of 8 be  $x$ . Then by the formula  $A = \frac{\sum xf}{\sum f}$

$$5 = \frac{1 \cdot 2 + 2 \cdot 3 + 4 \cdot 3 + 6 \cdot 2 + 8 \cdot x}{2 + 3 + 3 + 2 + x} = \frac{32 + 8x}{10 + x}$$

$$\text{or } 18 = 3x; \therefore x = 6$$

15. Ans. (B)

Sol. We have,  $(A \cup B) \cap B' = A$

$$[(A \cup B) \cap B'] \cup A' = A \cup A' = N$$

16. Ans. (A)

Sol. Here solution  $R$  is reflexive since  $1 + a \times a > 0 \forall$  real numbers  $a$ . It is symmetric since  $1 + ab > 0 \Rightarrow 1 + ba > 0$ . However  $R$  is not transitive : consider three real numbers  $2, -\frac{1}{6}$  and  $-2$ .

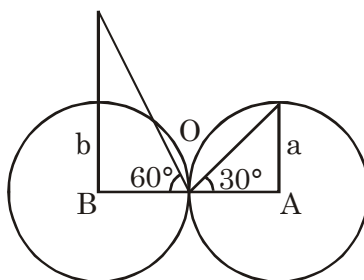
$$\text{We have } 1 + 2 \times \left(-\frac{1}{6}\right) = \frac{2}{3} > 0 \text{ and } 1 + \left(-\frac{1}{6}\right)(-2) = \frac{4}{3} > 0$$

$$\text{Hence, } 2R\left(-\frac{1}{6}\right) \text{ and } \left(-\frac{1}{6}\right)R(-2)$$

$$\text{But } 2R(-2) \text{ since } 1 + 2(-2) = -3 \not> 0$$

17. Ans. (B)

Sol. Let  $A$  and  $B$  be the centres of the two circles where the poles of height  $a$  and  $b$  respectively stand making angles  $30^\circ$  and  $60^\circ$  respectively at the point  $O$  where these circles touch each other externally.



Sum of the radii of the two circles =  $a \cot 30^\circ + b \cot 60^\circ = a\sqrt{3} + \frac{b}{\sqrt{3}} = \frac{3a+b}{\sqrt{3}}$   
 = distance between the centres of the two circles.

18. Ans. (C)

Sol.  $P(n) = 10^n + 3 \cdot 4^{n+2} + 5$  is divisible by 9.

For  $n = 1$ .

$$P(1) = 10^1 + 3 \cdot 4^{1+2} + 5$$

$$= 10 + 3 \times 3^3 + 5$$

$$= 10 + 192 + 5$$

$$= 207 \text{ is divisible by 9.}$$

Hence, the given statement is true for  $n = 1$ ,

i.e.,  $P(1)$  is true

Let the statement  $P(m)$  is true i.e.,

$$P(m) = 10^m + 3 \cdot 4^{m+2} + 5 \text{ is divisible by 9}$$

$$10^m + 3 \cdot 4^{m+2} + 5 = 9m \text{ (Let)}$$

Now, we have to prove that the given statement is true

for  $n = m + 1$ , i.e.,  $P(m + 1)$  is true.

$$\text{Now } 10^{(k+1)} + 3 \cdot 4^{(k+1)} + 2 + 5$$

$$P(m + 1) = 10^{m+1} + 3 \cdot 4^{(m+1)} + 2 + 5$$

$$= 10^m \cdot 10 + 3 \cdot 4^{m+2} \cdot 4 + 5$$

$$= (9m - 3 \cdot 4^{m+2} - 5) \cdot 10 + 3 \cdot 4^{m+2} \cdot 4 + 5$$

$$= 9 \cdot 10m - 30 \cdot 4^{m+2} - 50 + 12 \cdot 4^{m+2} + 5$$

$$= 9 \cdot 10m - 18 \cdot 4^{m+2} - 45$$

$$= 9[10m - 2 \cdot 4^{m+2} - 5] \text{ is divisible by 9}$$

Hence, the given statement is true for  $n = m + 1$ ,

i.e.,  $P(m + 1)$  is true.

Hence,  $P(n)$  is true for each natural number  $n \in \mathbb{N}$ .

Hence Proved.

19. Ans. (C)

Sol.

| p | q | $p \leftrightarrow q$ | $\sim q$ | $p \leftrightarrow \sim q$ | $\sim (p \leftrightarrow \sim q)$ |
|---|---|-----------------------|----------|----------------------------|-----------------------------------|
| T | T | T                     | F        | F                          | T                                 |
| T | F | F                     | T        | T                          | F                                 |
| F | T | F                     | F        | T                          | F                                 |
| F | F | T                     | T        | F                          | T                                 |

20. Ans. (C)

$$\text{Sol. } \bar{x}_1 = 4, N = 5 \text{ and } \frac{\sum (x - \bar{x})^2}{N} = 5.2 \Rightarrow \sum (x - \bar{x})^2 = (5.2)5$$

$$\therefore \sum (x - \bar{x})^2 = 26$$

$$\therefore (1-4)^2 + (2-4)^2 + (6-4)^2 + (\alpha-4)^2 + (\beta-4)^2 = 26$$

where  $\alpha, \beta$  are the other two observations.

$$\therefore 9 + 4 + 4 + (\alpha-4)^2 + (\beta-4)^2 = 26$$

$$\therefore (\alpha-4)^2 + (\beta-4)^2 = 9$$

$$\text{Also, } \frac{1+2+6+\alpha+\beta}{5} = 4$$

$$\therefore \alpha + \beta = 20 - 9 = 11$$

Clearly 4, 7 only satisfy the above equation in  $\alpha, \beta$ . Hence required numbers are 4, 7.

## SECTION-II

1. **Ans. 0.11**

$$\text{Sol. } I_n = \int \tan^n x dx = \int \tan^{n-2} x (\sec^2 x - 1) dx = \frac{\tan^{n-1} x}{n-1} - I_{n-2}$$

$$I_n + I_{n-2} = \frac{\tan^{n-1} x}{n-1}$$

2. **Ans 0.00**

$$\text{Sol. } f(x) = \frac{x}{x^2 + 16}$$

$$f'(x) \geq 0 \Rightarrow f(x) \text{ is increasing in } [-4, 4]$$

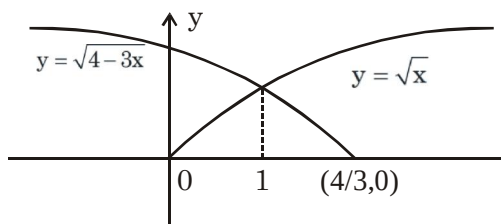
$$\text{minimum value is } f(0) = 0$$

$$\text{maximum value of is } f(4) = \frac{1}{17}$$

$$f(x) \in \left[0, \frac{1}{17}\right]$$

3. **Ans. 0.88 or 0.89**

Sol.



$$A = \int_0^1 (\sqrt{x}) dx + \int_1^{4/3} \sqrt{4-3x} dx = \frac{8}{9}$$

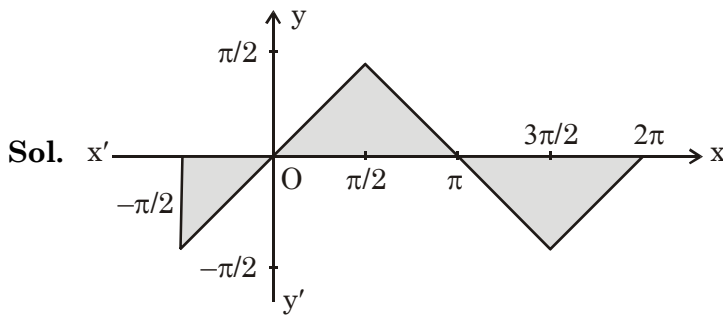
4. **Ans. 0.00**

$$\text{Sol. } 2(xdy + ydx) + ydy - dy = xdx + 5dx$$

$$2d(xy) + ydy - dy = xdx + 5dx$$

$$2xy + \frac{y^2}{2} - y = \frac{x^2}{2} + 5x + c$$

5. Ans. 0.25



$$\int_{-\pi/2}^{2\pi} \sin^{-1}(\sin x) dx = \underbrace{-\left(\frac{1}{2} \times \frac{\pi}{2} \times \frac{\pi}{2}\right)}_{\text{for } -\frac{\pi}{2} < x < 0} + \underbrace{\left(\frac{1}{2} + \pi \times \frac{\pi}{2}\right)}_{\text{for } 0 < x < \pi} - \underbrace{\left(\frac{1}{2} \times \pi \times \frac{\pi}{2}\right)}_{\text{for } \pi < x < 2\pi} = -\frac{\pi^2}{8}$$

6. Ans. 1.00

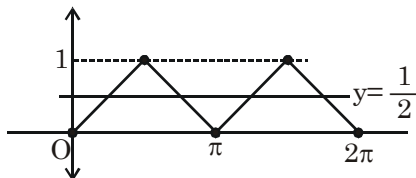
Sol.

$$\lim_{x \rightarrow 0} \frac{\int_0^x t^2 dt}{x - \sin x} = 2 \Rightarrow \lim_{x \rightarrow 0} \frac{\frac{x^3}{3}}{1 - \cos x} = 2 \quad (\text{by L.H. Rule})$$

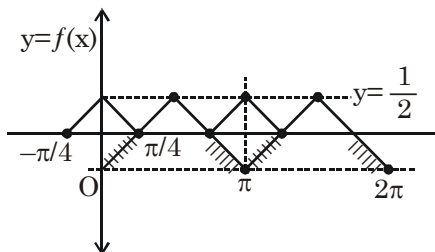
$$\Rightarrow \frac{2}{\sqrt{a}} = 2 \Rightarrow \boxed{a=1}$$

7. Ans. 3.75

Sol. Graph of  $y = g(x) = \left| \frac{2 \sin^{-1}(\sin x)}{\pi} \right|$



Fundamental period of  $g(x)$  is  $\pi$   
for fundamental of  $y = f(x) = |g(x) + a|$



$$\boxed{a = \frac{-1}{2}} \Rightarrow \int_a^0 \{x\} dx = \int_{-1/2}^0 (x+1) dx = 0.375$$

8. Ans. 0.50

Sol.  $f(x) = \int_0^x e^t \sin(x-t) dt$

let  $x - t = z$

$$f(x) = \int_0^x e^{x-z} \sin z \, dz = e^x \int_0^x e^{-z} \sin z \, dz$$

$$f'(x) = e^x \int_0^x e^{-z} \sin z \, dz + e^x + e^{-x} \sin x = f(x) + \sin x$$

$$\Rightarrow f'(x) - f(x) = \sin x$$

$$\Rightarrow f'\left(\frac{\pi}{4}\right) - f\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$$

9. Ans. 50.00

Sol. Let  $g(x) = \{x\} \times \left\{\frac{-x}{2}\right\}$

which is discontinuous at all odd integers &  $h(x) = \sin \pi x$ ,

• At  $x = 2m$ ,  $m \in \mathbb{I}$

$$h(2m) = \sin 2m\pi = 0$$

$\Rightarrow f(x)$  is differentiable at  $x = 2m$

• At  $x = 2m + 1$ ,  $m \in \mathbb{I}$ ,  $g(x)$  is discontinuous

$$h(2m + 1) = 0 \text{ but } h'(2m + 1) \neq 0,$$

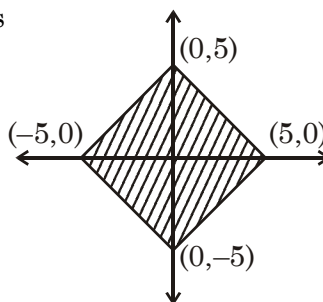
so  $f(x)$  is non differentiable at

$$x = 2m + 1, m \in \mathbb{I}$$

So  $n = 25$

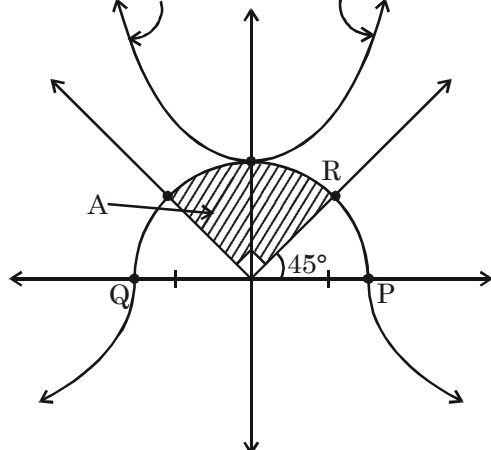
$$|y| + |x| \leq 5$$

$$\text{Area} = 50$$



10. Ans. 1.00

Sol.  $y|y| + x|x| = 4$   $y|y| - x|x| = 4$



$$\text{Required Area} = A = \frac{1}{4} \times \pi \times 4 = \pi$$

(since curve  $\widehat{PRQ}$  is circle with radius 2)

$$\text{So } \frac{A}{\pi} = 1$$